

Computing Market Tightness from the AD and AS Curves

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Goal : determine tightness $x \rightarrow$ from x we can infer all other variables in the model.

Household's spending/purchasing decision (by maximizing utility)

$$\underline{y = y^d(x, p^n(x))}$$

Household's visit/shopping decision (by matching process)

$$v = y / q(x)$$

$$y = v * q(x)$$

by definition : $x = v / k \Rightarrow v = x * k$

$$\Rightarrow y = x * q(x) * k$$

link b/w trading proba. $f(x) = x * q(x)$
by definition of probabilities (accounting) ordering $\uparrow$ buying $\uparrow$

$$\Rightarrow y = f(x) * \underline{k} = \underline{y^s(x) = y}$$

So in the model we always have:

$$y^d(x, p^n(x)) = y^s(x)$$

households
choose consumption
to max. utility

trades are governed
by matching function,
& capacity supplied

Tightness α is computed by solving
the $AD(\alpha) = AS(\alpha)$ equation

But need to specify a price norm $p^n(\alpha)$
first $\rightarrow$ we obtain $\neq \alpha$, and $\neq$
properties, for different price norms.